

PROJECT #3 - 2

Measure absolute speed from feature motion

Tracking billiard balls

1. Use video (even if it is pool): <https://youtu.be/Z9MgOKp5pbs?t=109>
 - <https://cutter.tube/> or instructions at <https://ytcutter.link/>
2. Object tracking: YOLO w/ YOLO26n.pt
 - Box balls, find out how centers of boxes move
 - Display trajectories
3. Measure and express speed of (some) balls in pixel/s
 - How noisy is it?
4. Recalibrate pixels to real space based on table dimensions:
 - table: 50 inches wide by 100 inches
 - ball diameter: 2.25 inches
 - Express speed of balls in m/s.
 - Express off-centeredness of collisions
5. Improvements (can be skipped FIRST):
 - Transform picture to a 1:2 rectangle before tracking
 - Use YOLO boxes as ROI for OpenCV HoughCircles to determine ball position as precisely as possible
6. Compare against physics:
 - Point model: are outgoing velocities perpendicular to each other?
 - Is kinetic energy conserved?